

DEB Installation for Windows 11

These instructions cover the software installation and configuration required for DEB participants.

[Section 1 – Installation of Python and 3rd party software](#)

[Section 2 – Configuration for DEB image capture and processing](#)

[Section 3 – Set up and configuration of DEB upload tools](#)

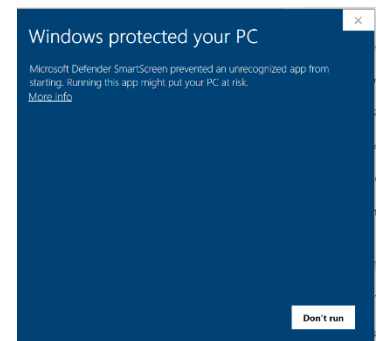
[Appendix I – Converting existing \(2025 or earlier\) DEB installations to the 2026 configuration](#)

Section 1 – Python and 3rd Party Software Installation

Follow these procedures to install the 3rd party software for the DEB project. You can skip any step that you have already completed.

You may need to have some of these websites whitelisted by your institution to download and install some programs.

Several of these installs may cause the Windows Protect to come up. Select **More Info** (Right image) and you can then select to run the installation.



STEP 1a – PYTHON INSTALLATION I

We will install two versions of Python in this guide.

ADVANCED USERS: *This procedure installs Python >=3.13 to PATH, as it is deemed the simplest method for novice users. Feel free to alter the configuration if you know what you are doing. The instructions will reference Python >=3.13 as python and Python 3.8 as path/to/Python38/python*

Follow instructions (Right) to check for existing Python installations.

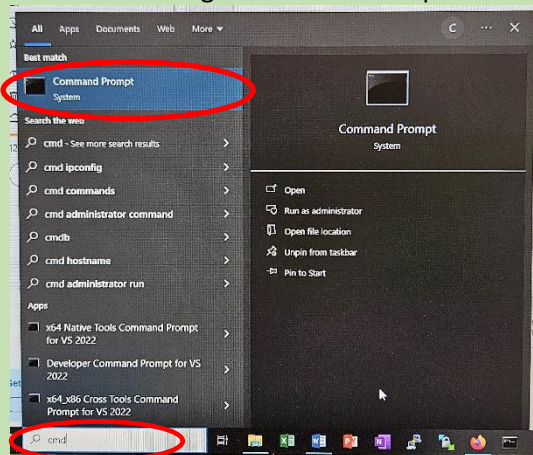
If Python is found and is 3.12 or higher, you can skip to [STEP 1b – PYTHON INSTALLATION II](#)

If Python is found but is 3.11 or lower, you may want to remove Python from PATH or delete it. Contact the DEB coordinator for assistance with this decision if you need it. You will still need both Python 3.8 and 3.13 for this system.

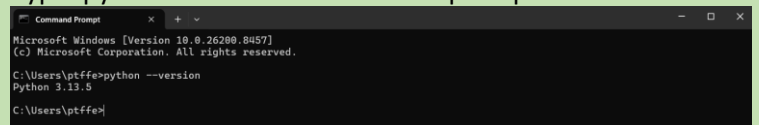
If no Python found: continue

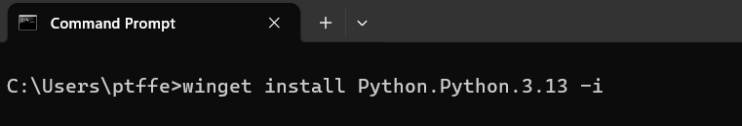
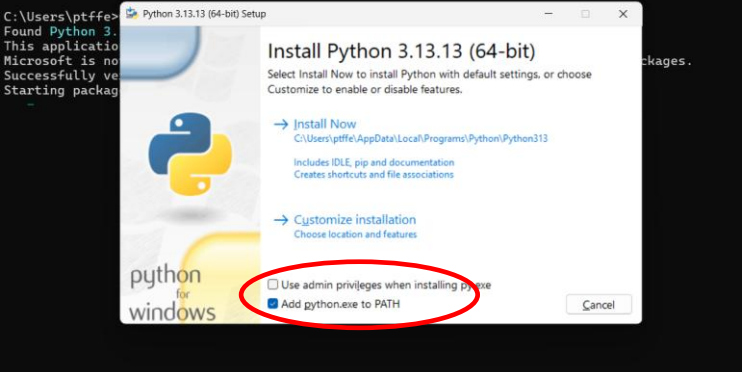
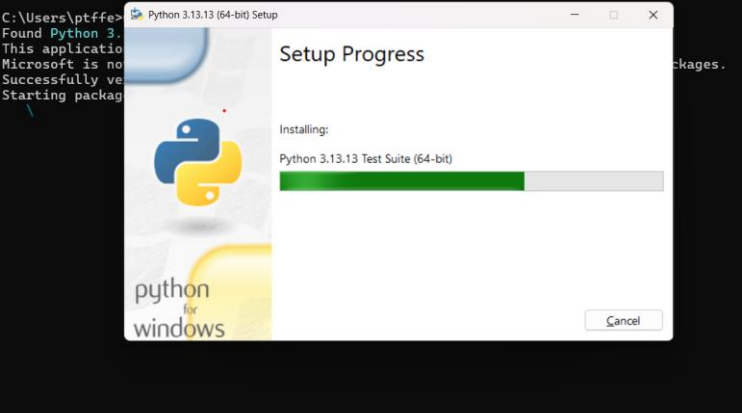
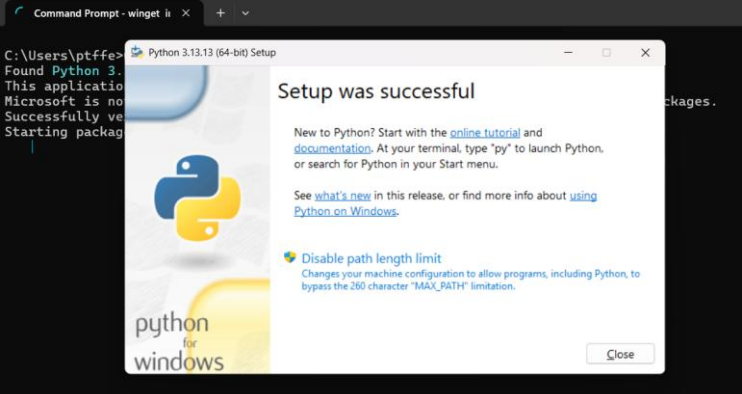
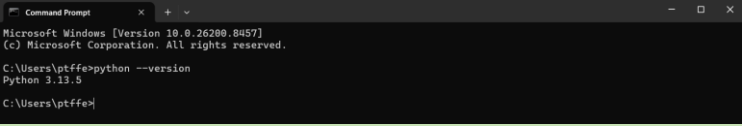
CHECK FOR EXISTING PYTHON!

Open Command Prompt by typing 'cmd' in Windows search bar and selecting 'Command Prompt'

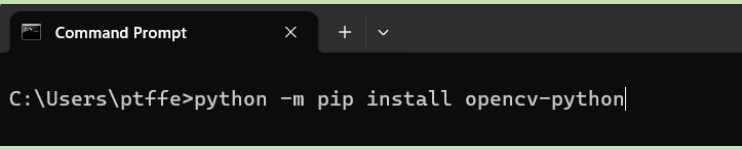
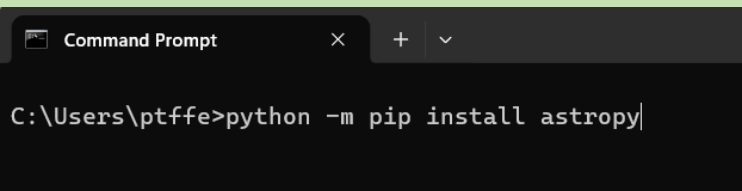
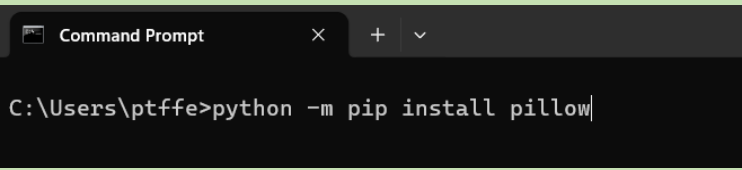
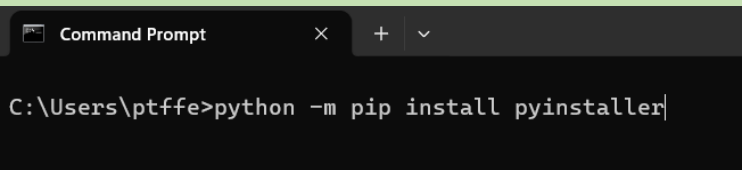
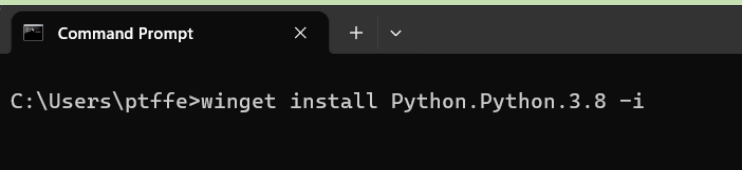


Type 'python --version' in command prompt

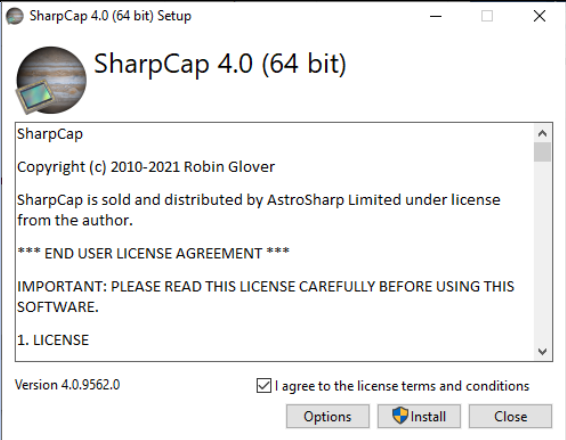
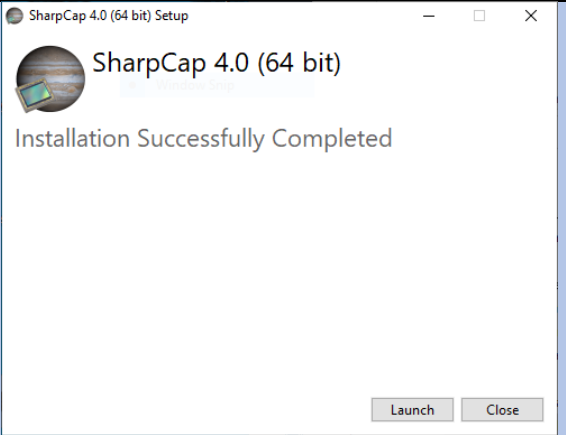
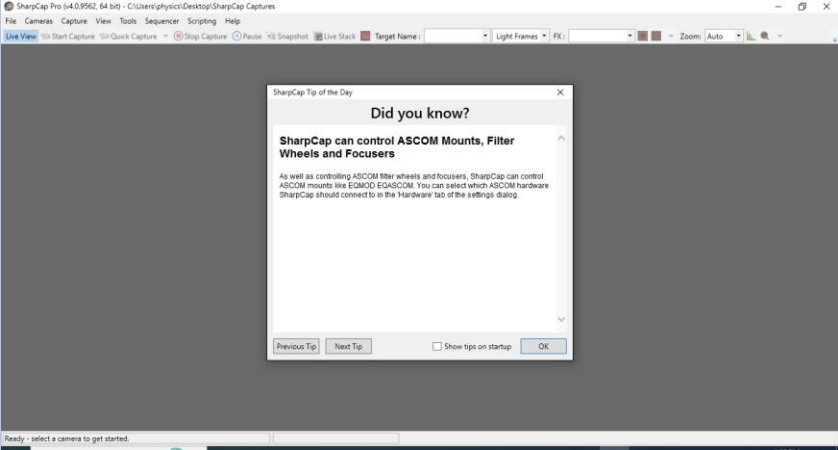


We will use Windows built in winget to install Python Enter the following command at the cmd prompt: winget install Python.Python.3.13 -i	Type 'winget install Python.Python.3.13 -I' in cmd prompt 
The installer will appear as image to the right. <u>Verify "Add python.exe to PATH" is checked.</u> Select "Install Now"	
Installer will display installation progress	
When installation complete select "Close"	
Running the "python --version" command should now show a Python 3.13 installation. If not, enter: winget uninstall Python.Python.3.13 Restart computer Repeat steps above.	Type 'python --version' in command prompt 

STEP 1b – PYTHON INSTALLATION II

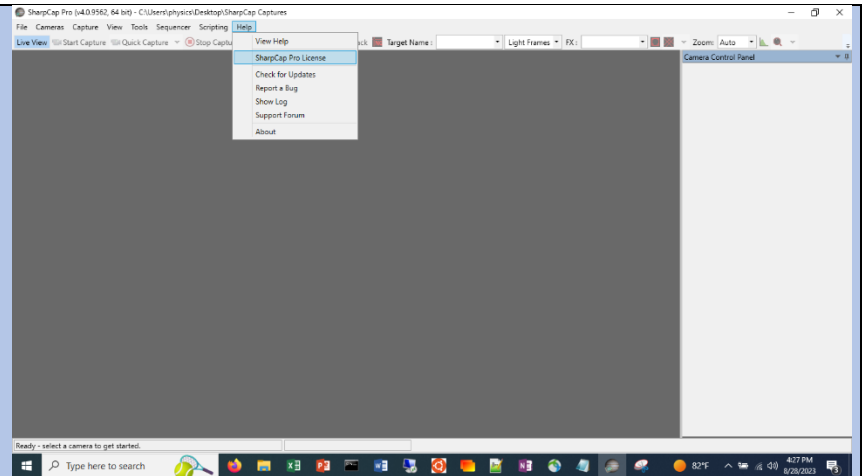
Install Python 3.13 packages required for processing Install cv2 in the command prompt with: python -m pip install opencv-python	
Install astropy in the command prompt with: python -m pip install astropy	
Install PIL in the command prompt with: python -m pip install pillow	
Install pyinstaller in the command prompt with: python -m pip install pyinstaller	
Install Python 3.8 for processing Install Python 3.8 with winget from command prompt with the following command: winget install Python.Python.3.8 -i Write down the install path for later reference. This will be referred to as path\to\Python38 later in the instructions IMPORTANT: Verify that “Add Python 3.8 to PATH” box is <u>NOT CHECKED</u> This will progress just like the previous Python installation, so no more slides will be shown for this process.	 

STEP 2 – SHARPCAP INSTALL

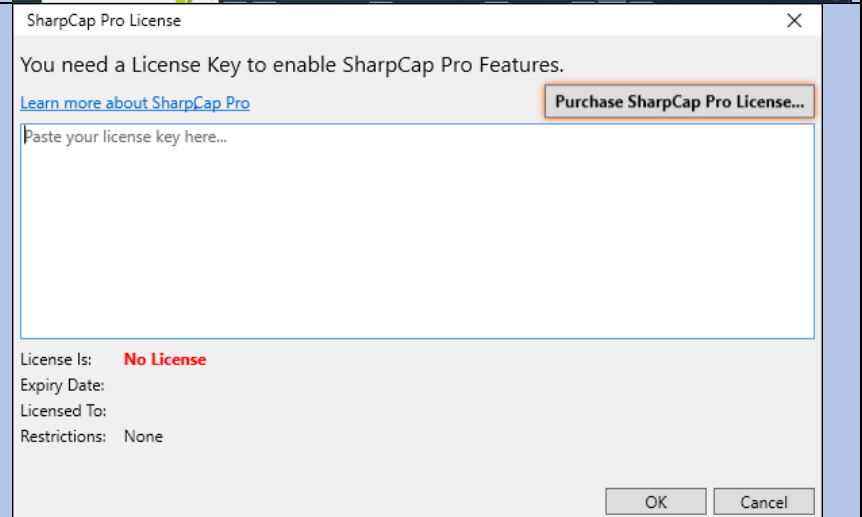
Install SharpCap 4.1 (64 bit)	Downloads – SharpCap – Lunar, Planetary, Solar and Deep Sky Imaging. EAA and Live Stacking.	
Agree and install		
Click Launch		
The program will open with an initial information panel that you must close. (image of information panel not included)		

Add Pro License

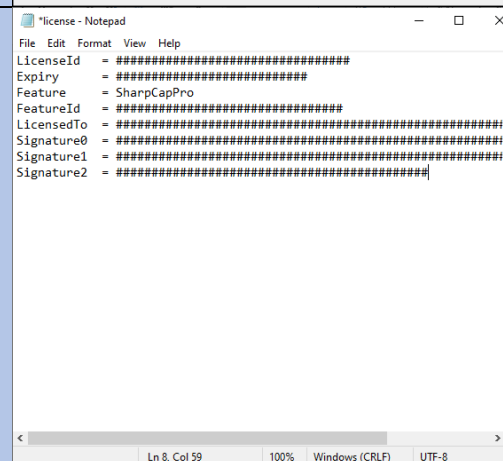
Select 'Help' and 'SharpCap Pro License'



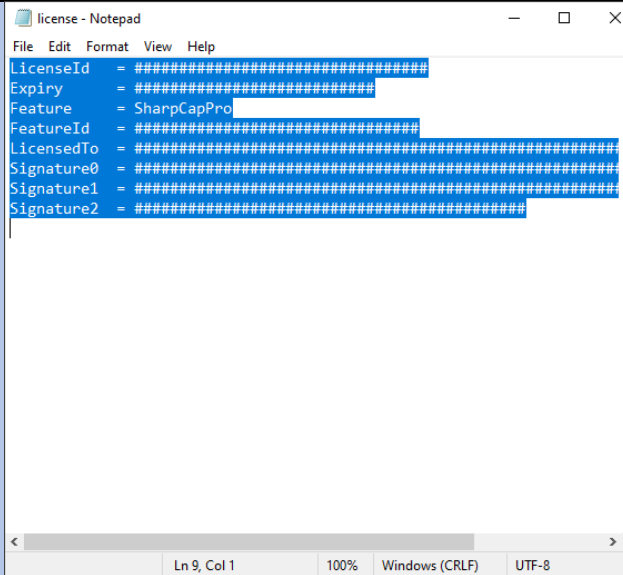
The License Key window will then open.



You will need to request the license.txt file from your DEB contact



Copy the data in the license.txt file by clicking “control” + “a” and then “control” + “c” in the text file while it is open in the Notepad text editor.

A screenshot of a Notepad window titled "license - Notepad". The menu bar includes File, Edit, Format, View, and Help. The text content is as follows:

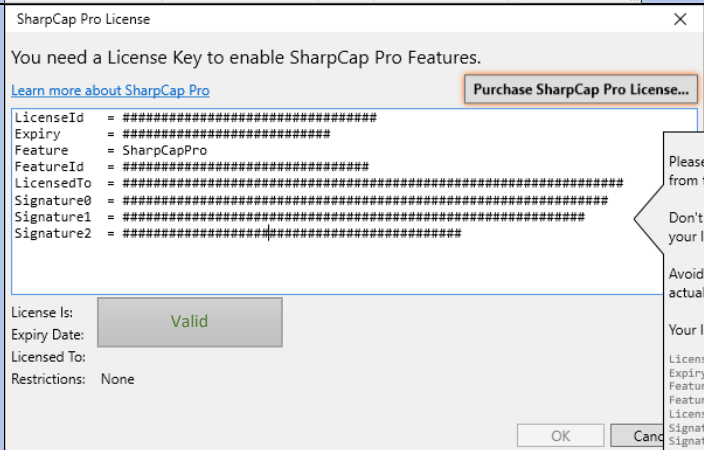
```
LicenseId = #####  
Expiry = #####  
Feature = SharpCapPro  
FeatureId = #####  
LicensedTo = #####  
Signature0 = #####  
Signature1 = #####  
Signature2 = #####
```

The status bar at the bottom shows "Ln 9, Col 1", "100%", "Windows (CRLF)", and "UTF-8".

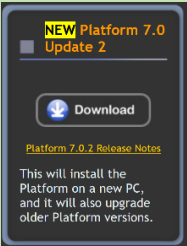
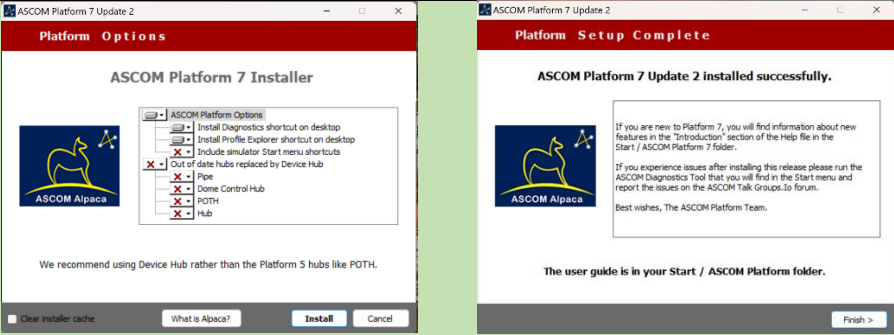
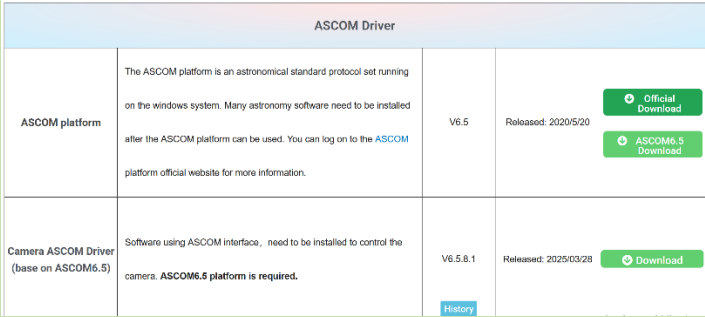
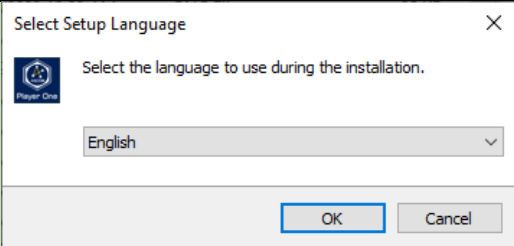
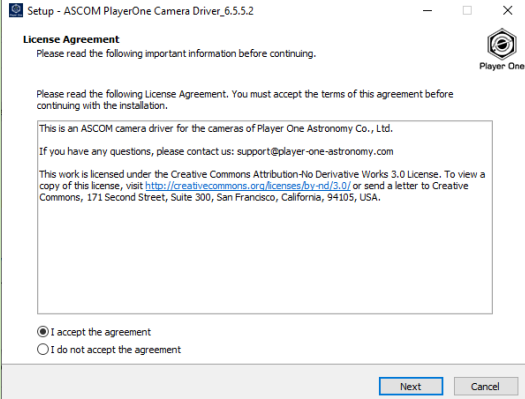
Go back to SharpCap and use “control” + “v” to paste the license information into the box.

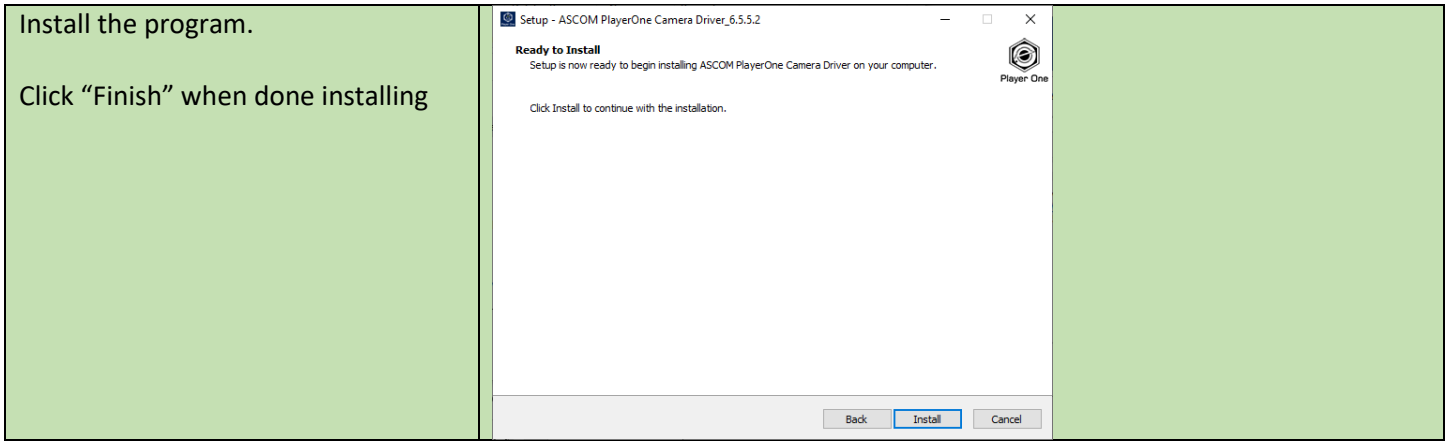
Click ok and then you should get a message that the license is valid.

You can close SharpCap now.

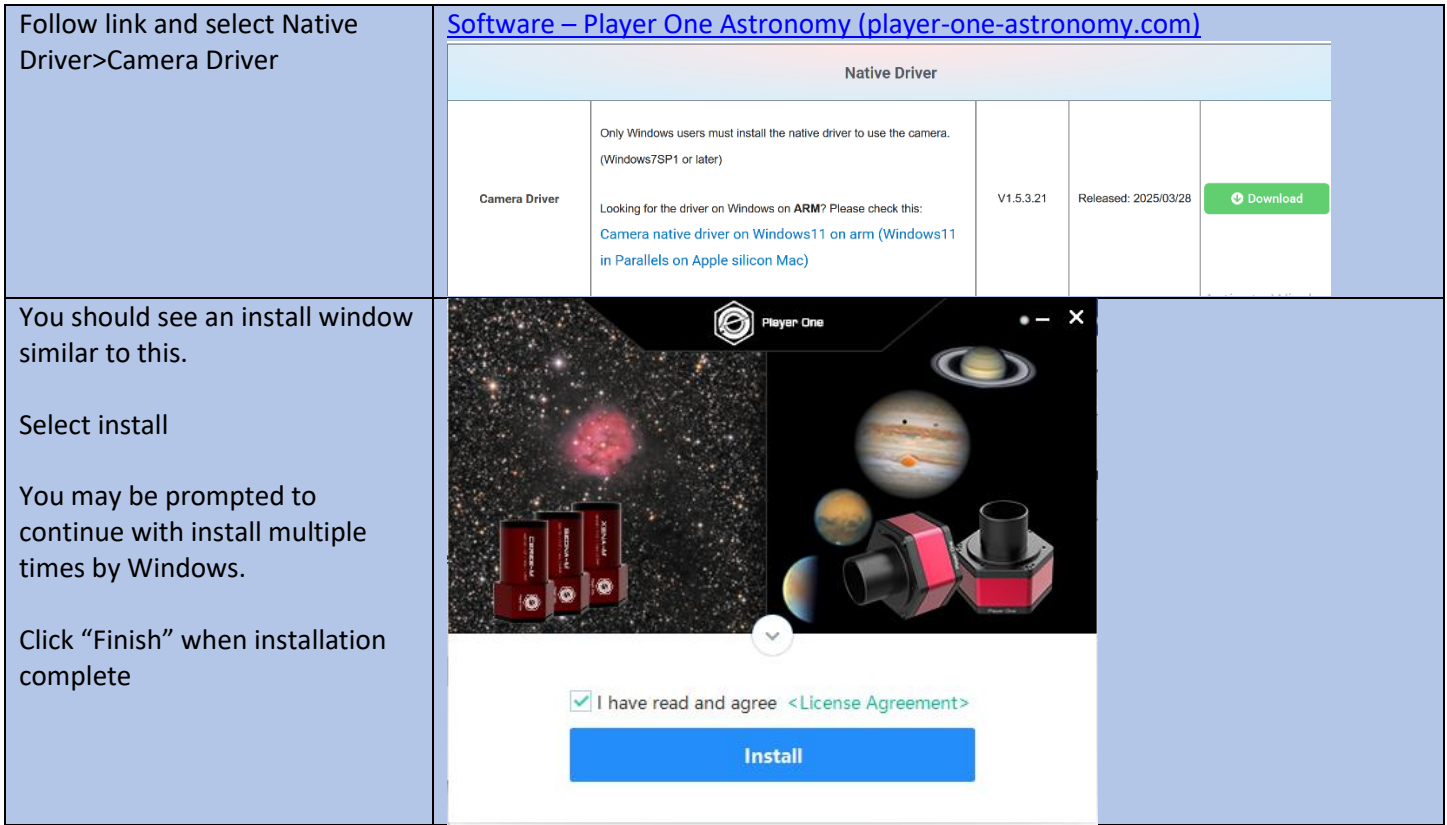
A screenshot of the "SharpCap Pro License" dialog box. The title bar says "SharpCap Pro License" with a close button. The main text says "You need a License Key to enable SharpCap Pro Features." Below this is a link "Learn more about SharpCap Pro" and a button "Purchase SharpCap Pro License...". A text box contains the same license information as the Notepad window. Below the text box, it says "License Is: Valid" with a green "Valid" button. Other fields include "Expiry Date:", "Licensed To:", and "Restrictions: None". At the bottom are "OK" and "Cancel" buttons. On the right side, there is a vertical list of license-related terms: "Please from t", "Don't your l", "Avoid actual", "Your l", "Licen", "Expiry", "Featur", "Featur", "Licen", "Signat", "Signat".

STEP 3 – ASCOM PlayerOne Camera Setup

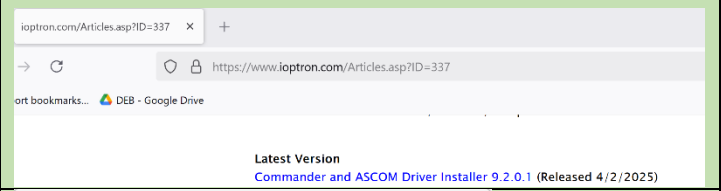
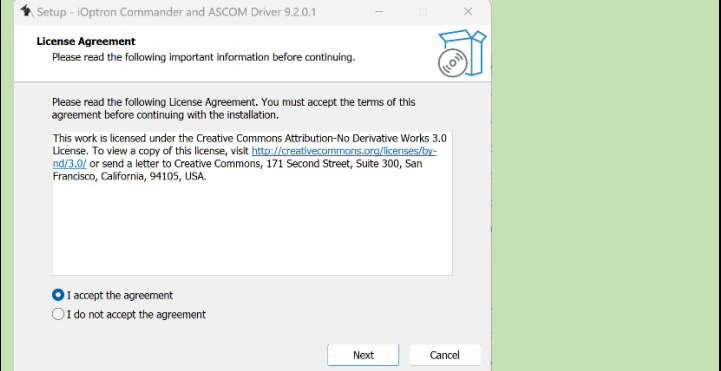
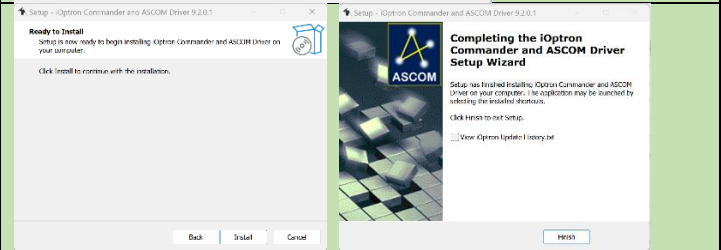
You must have the ASCOM platform installed first. Follow the link and click on the Download button in the window most similar to the one shown here.	Download Center (ascom-standards.org) 
Follow the instructions and use the default settings to install.	
Install ASCOM PlayerOne Camera Driver Open the provided installer or follow the link and select the “Camera ASCOM Driver...” version shown here	Software – Player One Astronomy (player-one-astronomy.com) 
Select Setup Language	
Accept the agreement	



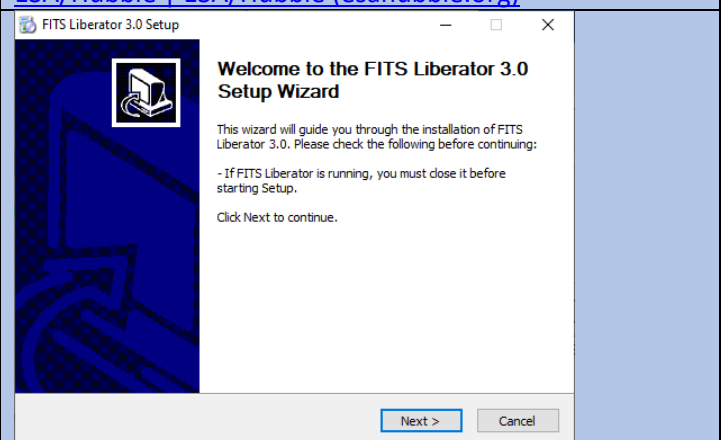
STEP 4 – Player one camera driver install

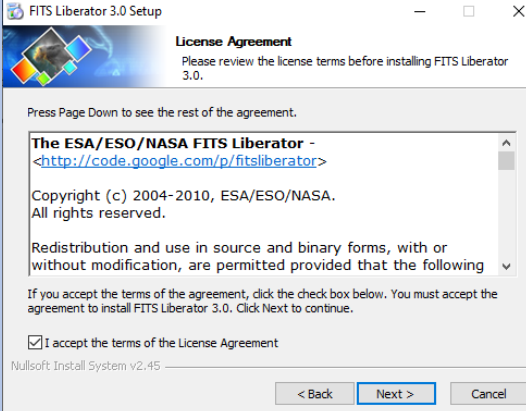
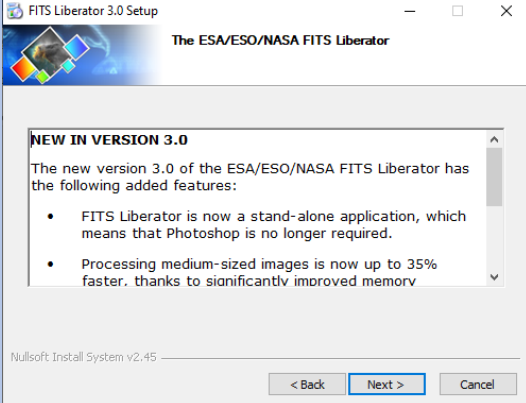
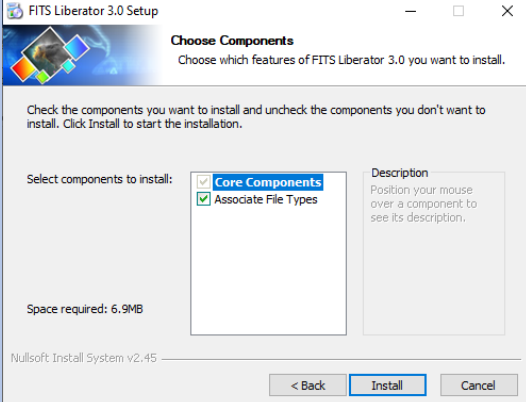


STEP 5 – iOptronCommander ASCOM Driver

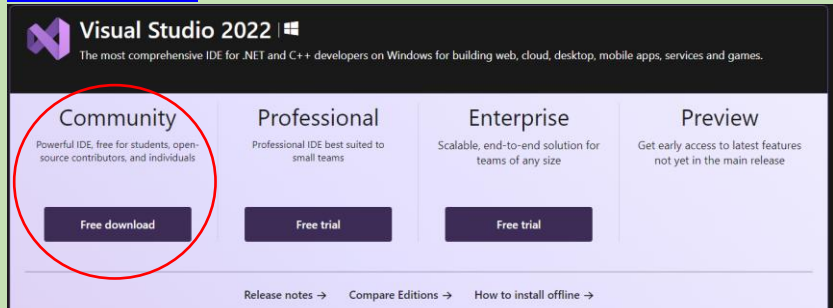
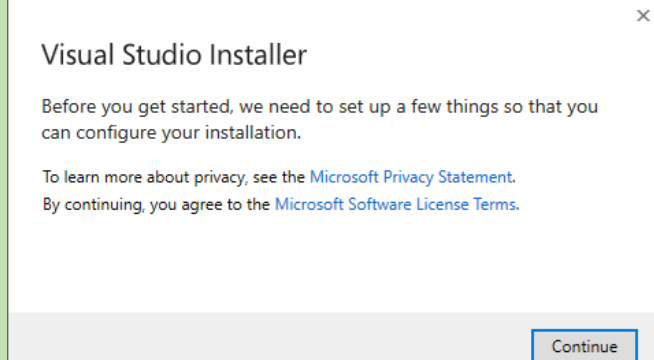
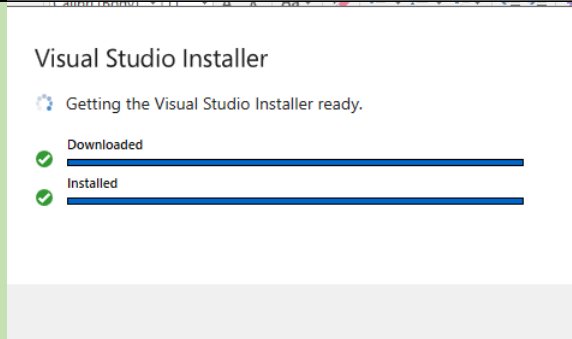
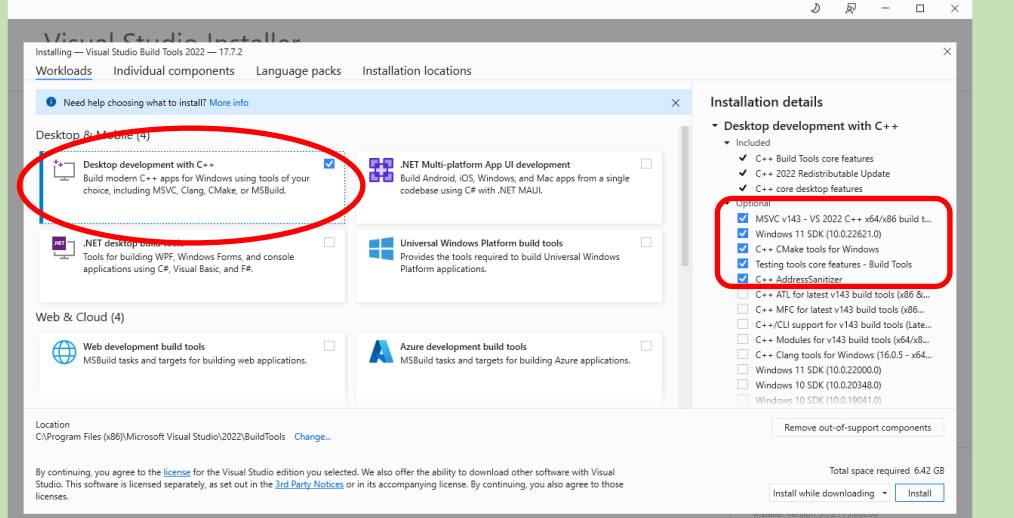
Follow link and select “Latest Version > Commander and ASCOM Driver Installer x.x.x.x”	ioptron.com/Articles.asp?ID=337 
Accept the License Agreement	
Select install and it will run.	

STEP 6 – FITS Liberator 3.0 (this is FITS file viewing program from ESO/ESA/NASA)

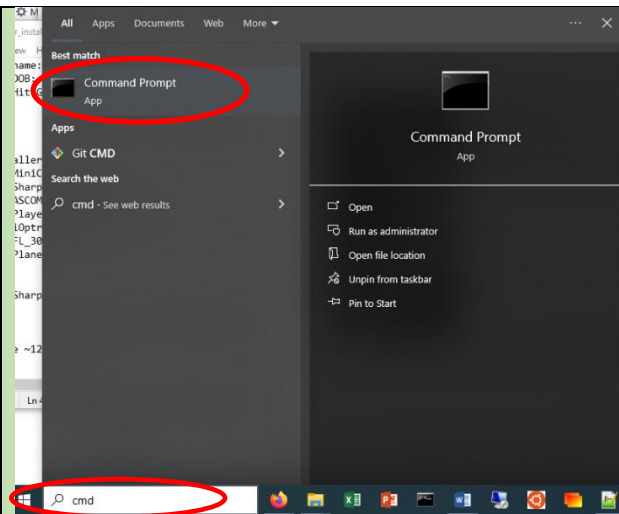
Follow link to download	Download the ESA/ESO/NASA FITS Liberator v3.0 ESA/Hubble ESA/Hubble (esahubble.org)
Download and run the setup file.	

Accept license agreement terms	 The screenshot shows the 'License Agreement' window for FITS Liberator 3.0. It contains the text of the license agreement, including copyright information (© 2004-2010, ESA/ESO/NASA) and redistribution terms. At the bottom, there is a checkbox labeled 'I accept the terms of the License Agreement' which is checked. Navigation buttons '< Back', 'Next >', and 'Cancel' are visible.	
Select next.	 The screenshot shows the 'NEW IN VERSION 3.0' window. It lists new features for version 3.0, such as it being a stand-alone application and faster processing. At the bottom, the 'Next >' button is highlighted in blue.	
Make sure both boxes are checked. Click install.	 The screenshot shows the 'Choose Components' window. It allows users to select which features to install. Under 'Select components to install:', both 'Core Components' and 'Associate File Types' are checked. A 'Description' box on the right provides more details. At the bottom, the 'Install' button is highlighted in blue.	

STEP 7 – Install Planetary System Stacker with Python

Follow link and select the “Community” > Free download Run installer	Download Visual Studio Tools - Install Free for Windows, Mac, Linux (microsoft.com)  The screenshot shows the Visual Studio 2022 download page. The 'Community' edition is highlighted with a red circle, and its 'Free download' button is also circled in red. Other editions like Professional, Enterprise, and Preview are visible with 'Free trial' buttons.
Install C++ build tools with vs_BuildTools (you might not see this window)	 The screenshot shows the 'Visual Studio Installer' window. It contains a message: 'Before you get started, we need to set up a few things so that you can configure your installation.' Below this is a 'Continue' button.
The installer may take some time to install.	 The screenshot shows the 'Visual Studio Installer' window with progress bars. The 'Downloaded' and 'Installed' sections both show 100% completion with green checkmarks.
Scroll down to the Desktop and Mobile section. Select the “Desktop development with C++” as seen here. You need the 5 blue checked boxes (far right of image) Click install (Bottom Right). This will take several minutes to install.	 The screenshot shows the 'Visual Studio Build Tools' installation window. The 'Desktop & Mobile (4)' section is selected and circled in red. In the 'Installation details' pane on the right, five components are checked with blue boxes: 'MSVC v143 - VS 2022 C++ x64/x86 build tools', 'Windows 11 SDK (10.0.22621.0)', 'C++ CMake tools for Windows', 'Testing tools core features - Build Tools', and 'C++ AddressSanitizer'.

Open a command prompt just like you did in Step 1.



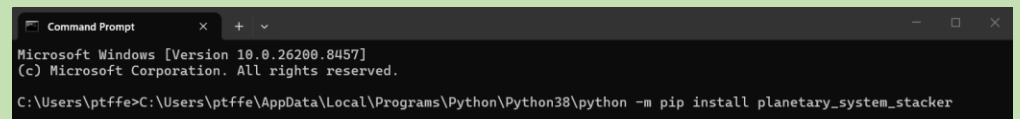
If you have installed python with these instructions, you will now use the path\to\Python38 you wrote down in Step 1b

If you forgot to write it down, check the following location for the Python38 folder:
C:\Users\yourUserName\AppData\Local\Programs\Python\

Install Planetary System Stacker with python.

Type: path\to\Python38\python -m pip install planetary_system_stacker

This might take a few minutes.

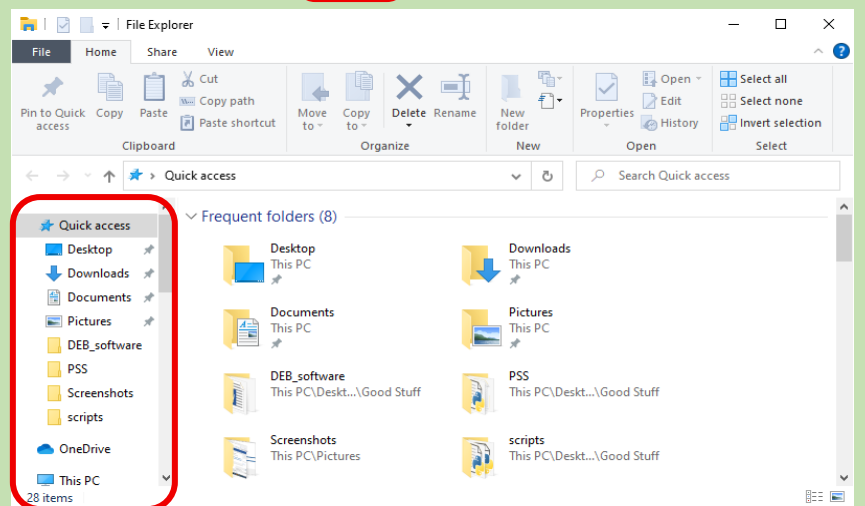


Go to Windows File Explorer icon at the bottom of your computer screen and move to the folder containing planetary_system_stacker.py

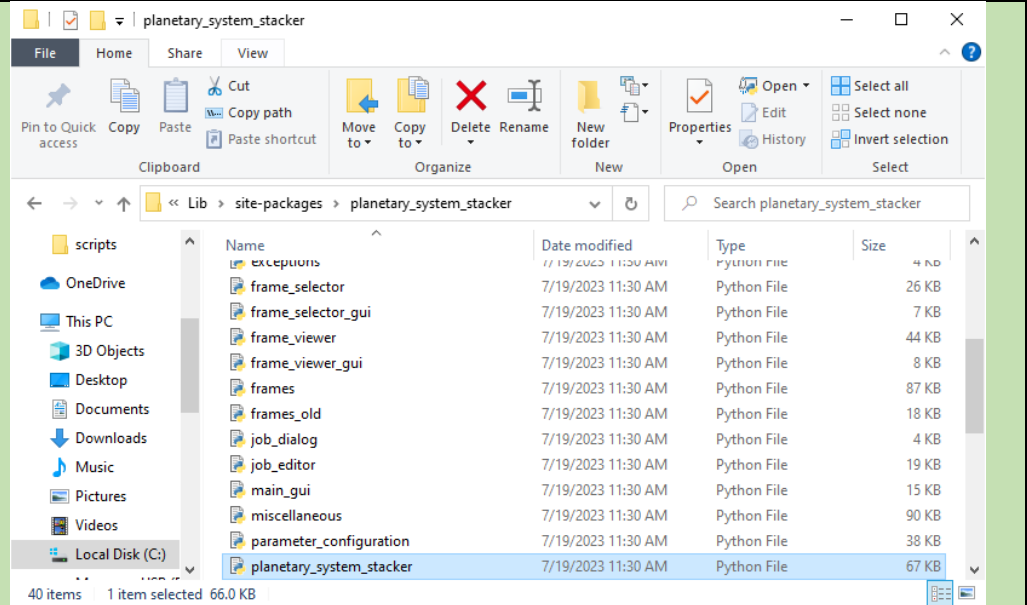


This will be in the path\to\Python38\Lib\site-packages\planetary_system_stacker folder

You can use the left navigation panel to find the folder; starting with "OS (C:)"



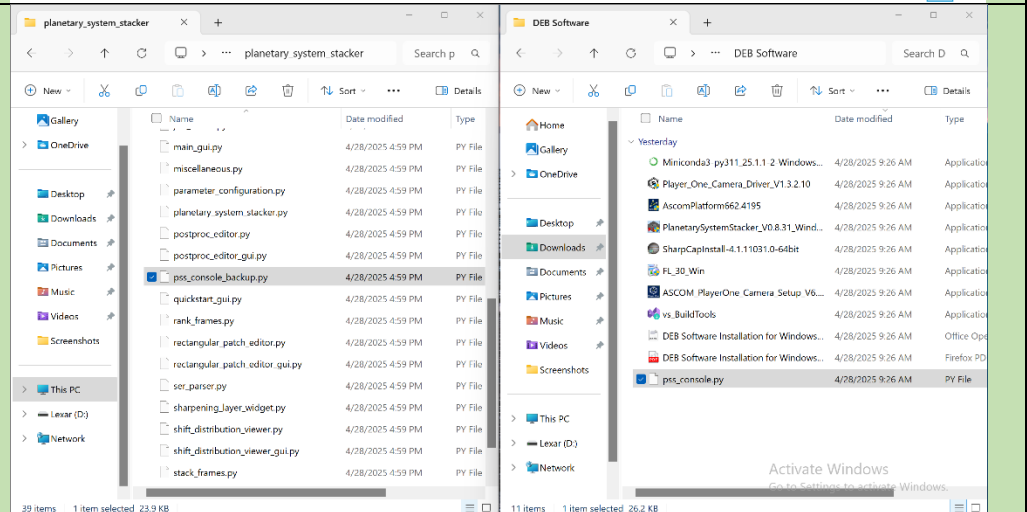
You should see a screen similar to this when you have found the installation folder



Replace pss_console.py in this folder:

- 1) Change the name of the existing pss_console.py to pss_console_backup.py
- 2) Copy or move the **pss_console.py** you downloaded from DEB into this location

RESTART COMPUTER



Congratulations!!

You have successfully installed all of the required 3rd party software.

Section 2 – Setup DEB Image Capture Configuration

The DEB configuration functions best when all of the code is in the same folder on your computer, and it is often convenient to have all DEB subfolders in a main DEB folder.

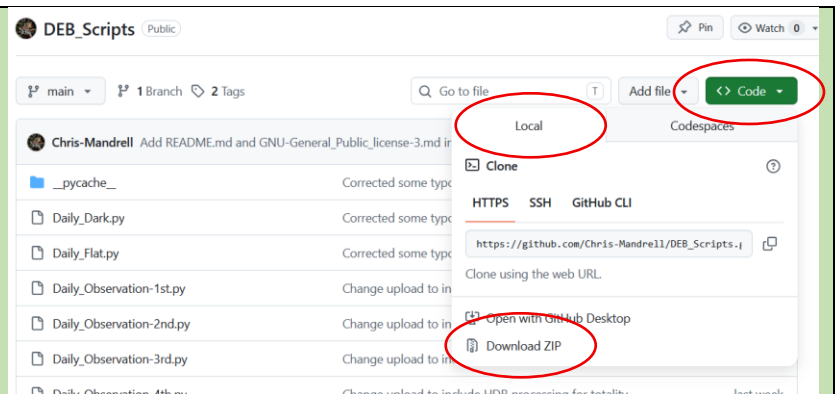
The default DEB folder for Python code is **C:\DEB\Programs**, but you can put the code in any folder you prefer. These instructions will also walk you through the process of specifying a folder other than the default.

You should create the main DEB folder and the Programs folder now. If you need assistance with this, a Google search will explain better than I could here how to add new folders.

Download the DEB scripts from:
https://github.com/Chris-Mandrell/DEB_Scripts

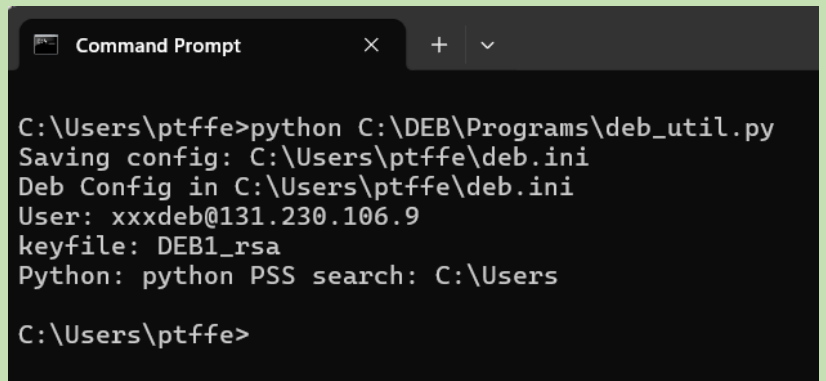
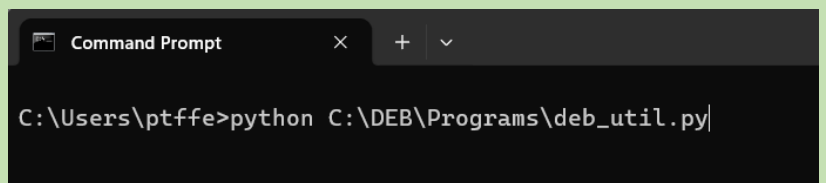
- Select <>Code (green button)
- Select Local tab
- Select Download ZIP

Unzip and move or copy/paste to the Programs folder you created (as described above)



Open a command prompt and run the following command with the proper path for your programs folder (default shown below):
`python C:\DEB\Programs\deb_util.py`

You will see something like this when done.



The output from deb_util.py provides us with the location of the deb.ini file in the “Saving config:” line

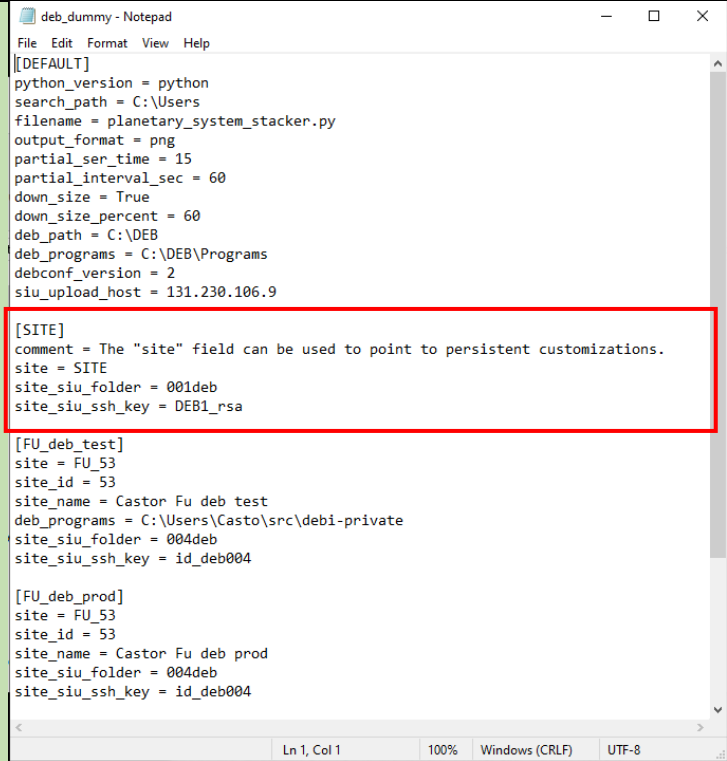
We need to alter that file to optimize some of the code.

Any changes we make to this file must be in the [SITE] section. Changes to the [DEFAULT] section will be reset every time any of the code runs.

We will add search_path and python_version variables to the [SITE] section for planetary system stacker and Python 3.8 respectively

The “site_siu_folder” and “site_siu_ssh_key” variables will be addressed later in the instructions

The next slide shows these changes



```
deb_dummy - Notepad
File Edit Format View Help
[DEFAULT]
python_version = python
search_path = C:\Users
filename = planetary_system_stacker.py
output_format = png
partial_ser_time = 15
partial_interval_sec = 60
down_size = True
down_size_percent = 60
deb_path = C:\DEB
deb_programs = C:\DEB\Programs
debconf_version = 2
siu_upload_host = 131.230.106.9

[SITE]
comment = The "site" field can be used to point to persistent customizations.
site = SITE
site_siu_folder = 001deb
site_siu_ssh_key = DEB1_rsa

[FU_deb_test]
site = FU_53
site_id = 53
site_name = Castor Fu deb test
deb_programs = C:\Users\Casto\src\debi-private
site_siu_folder = 004deb
site_siu_ssh_key = id_deb004

[FU_deb_prod]
site = FU_53
site_id = 53
site_name = Castor Fu deb prod
site_siu_folder = 004deb
site_siu_ssh_key = id_deb004

Ln 1, Col 1 100% Windows (CRLF) UTF-8
```

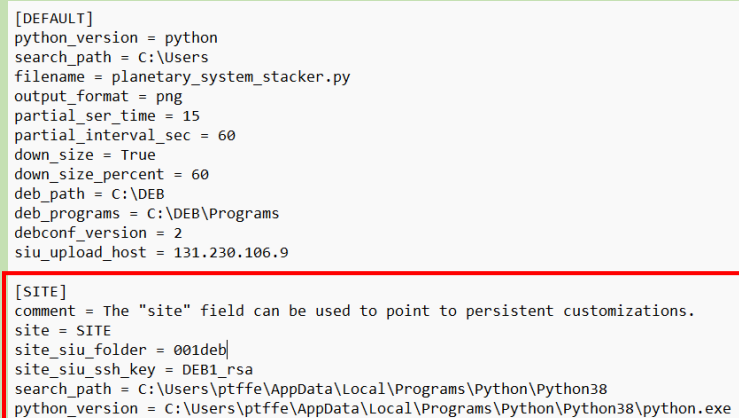
We will add:

```
search_path = path\to\Python38
python_version=path\to\Python38\python.exe
```

The example shows the authors specific locations for these variables. . .yours should match your system

If you put your programs in a location other than the default or want to capture to another folder, you should add a line to the [SITE] section with the correct address. Below is a list of [DEFAULT] variable purposes in case you need make changes to match your desires.

deb_path = location of image capture folder
deb_programs = location of programs folder



```
[DEFAULT]
python_version = python
search_path = C:\Users
filename = planetary_system_stacker.py
output_format = png
partial_ser_time = 15
partial_interval_sec = 60
down_size = True
down_size_percent = 60
deb_path = C:\DEB
deb_programs = C:\DEB\Programs
debconf_version = 2
siu_upload_host = 131.230.106.9

[SITE]
comment = The "site" field can be used to point to persistent customizations.
site = SITE
site_siu_folder = 001deb
site_siu_ssh_key = DEB1_rsa
search_path = C:\Users\ptffe\AppData\Local\Programs\Python\Python38
python_version = C:\Users\ptffe\AppData\Local\Programs\Python\Python38\python.exe
```

Congratulations!!

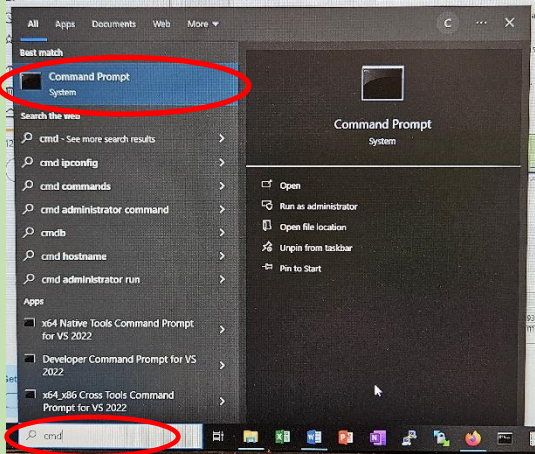
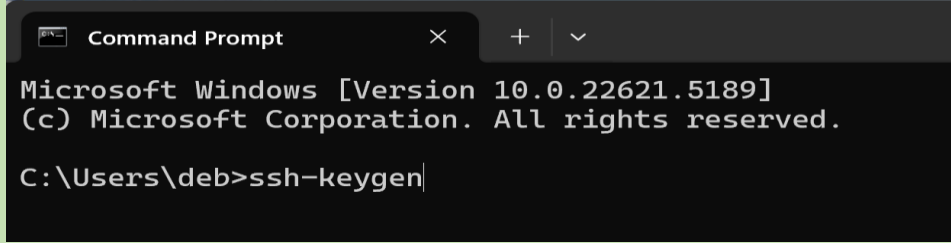
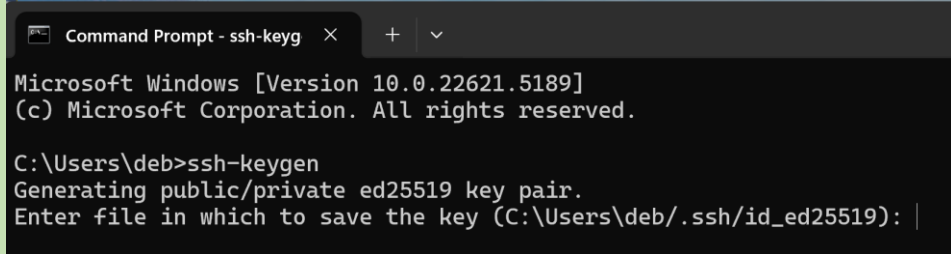
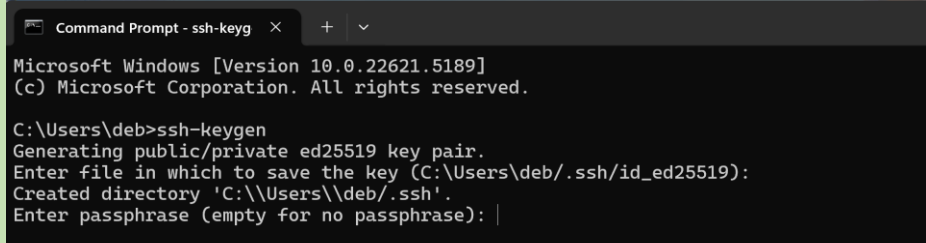
You have successfully setup the DEB capture/processing configuration.

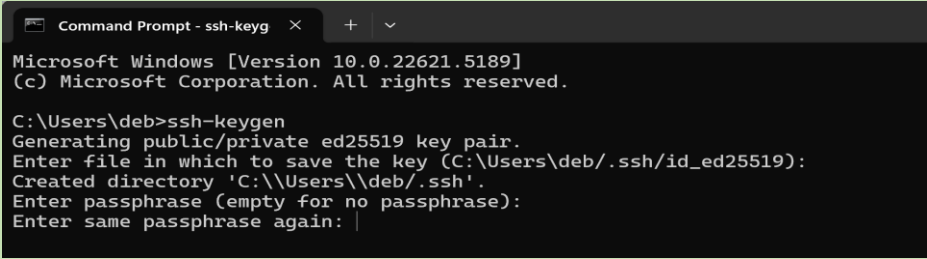
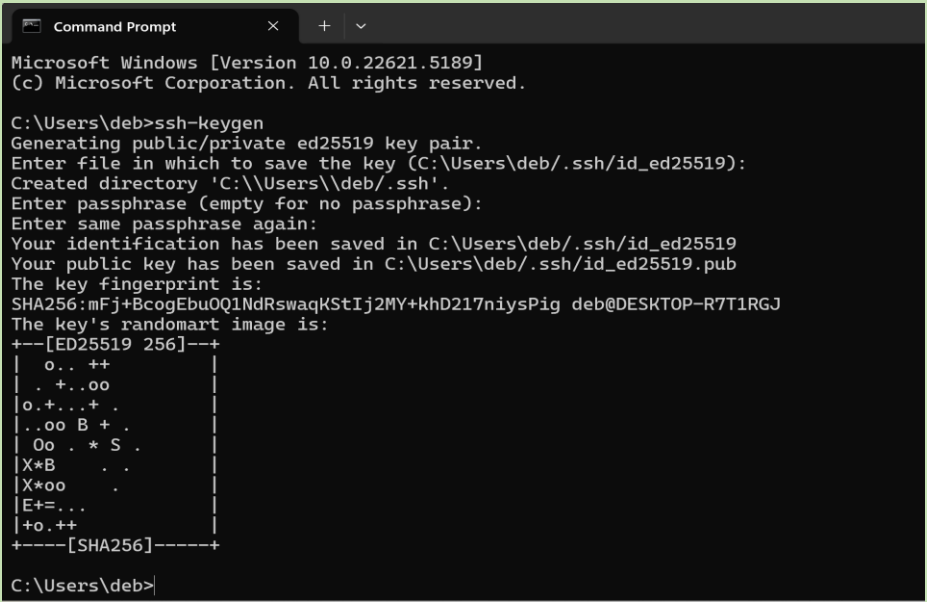
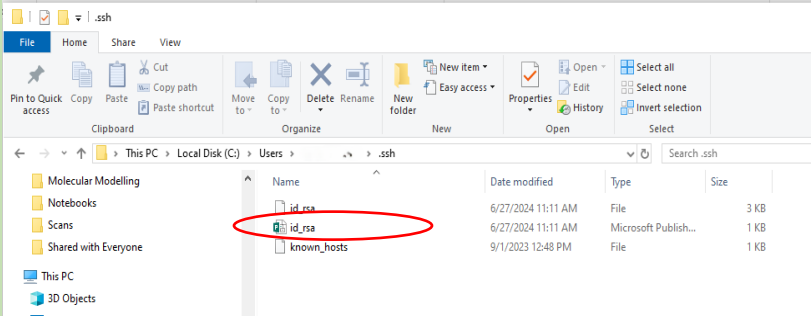
You can now image with the DEB software.

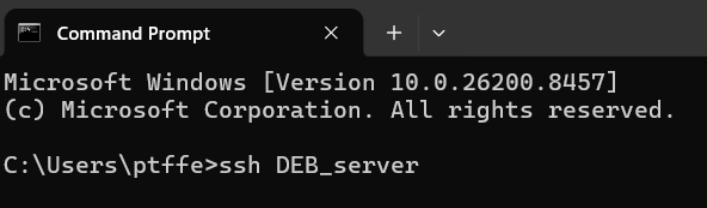
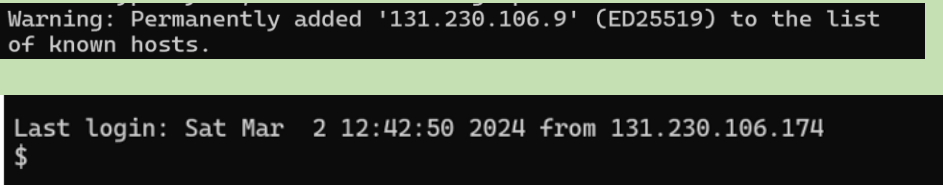
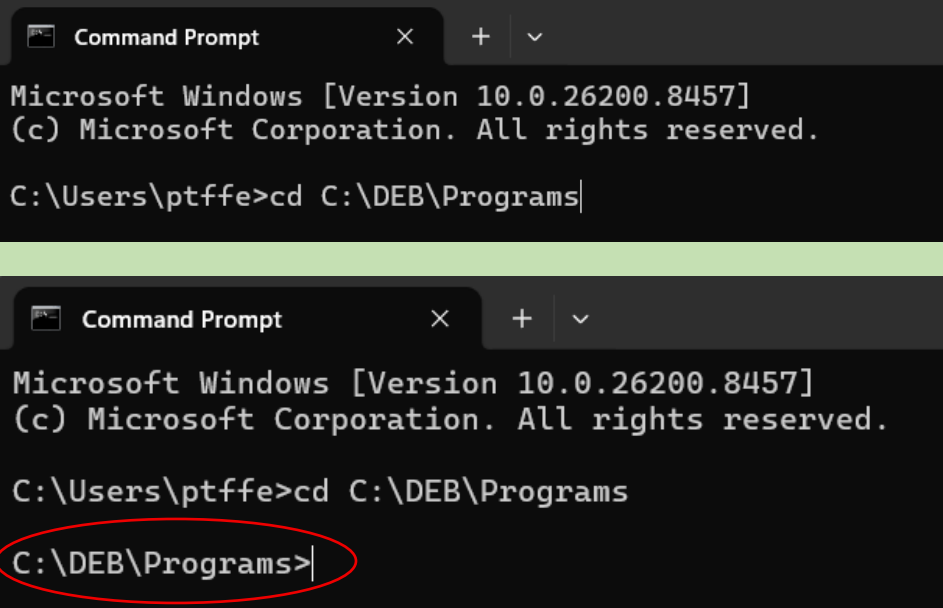
Section 3 – Setup/Configure DEB Uploading

The previous sections covered the steps to enable you to capture images with the DEB software, but DEB members will also upload images for reference and outreach.

This section covers the setup and configuration of these upload tools.

Open any command prompt	
Enter the following command to create an ssh key pair to securely connect to the DEB server: ssh-keygen	
We will use defaults throughout. Hit <enter> when prompted with any questions	
Be sure to only hit <enter> at passphrase prompt	

You have to <enter> again to confirm “no passphrase”	
Your screen should look something like this after completed.	
Use file explorer to navigate to: C:\Users\yourUsername\.ssh You will send the DEB coordinator a copy of your PUBLIC key. (your key will probably have a different name) If you send the private key, the DEB coordinator will require you to delete your key-pair and repeat the process.	
When the DEB coordinator sends confirmation that your key is in place, they will provide you with instructions on the next step.	Follow DEB coordinator’s instructions for next step.

Next, you will need to perform a “Handshake” with the server. After you have followed the coordinator’s instructions in the previous step, enter the following command at a command prompt: ssh DEB_server	 <pre> Command Prompt Microsoft Windows [Version 10.0.26200.8457] (c) Microsoft Corporation. All rights reserved. C:\Users\ptffe>ssh DEB_server </pre>
A successful “Handshake” will result in a “Warning” message before you are taken to the DEB server. You can look in your folder with the following list command: ls You can return to your command prompt with the following command to close the server: exit	 <pre> Warning: Permanently added '131.230.106.9' (ED25519) to the list of known hosts. Last login: Sat Mar 2 12:42:50 2024 from 131.230.106.174 \$ </pre>
We will now create a Windows executable from the upload.py program in your Programs folder. In any command prompt change to your Programs folder with: cd C:\DEB\Programs You will see a new location in your command prompt, if you weren’t already in the Programs folder.	Use the path to your Programs folder if you are not using default settings!  <pre> Command Prompt Microsoft Windows [Version 10.0.26200.8457] (c) Microsoft Corporation. All rights reserved. C:\Users\ptffe>cd C:\DEB\Programs Command Prompt Microsoft Windows [Version 10.0.26200.8457] (c) Microsoft Corporation. All rights reserved. C:\Users\ptffe>cd C:\DEB\Programs C:\DEB\Programs> </pre>

Use pyinstaller to create executable with the following command:
python -m PyInstaller upload.py

If it already exists, you will be asked if you want to overwrite it. Answer yes.

The results will end with something like the image to the right.

The last line tells us where the executable is located.

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ptffe>cd C:\DEB\Programs

C:\DEB\Programs>python -m PyInstaller upload.py|
```

```
11922 INFO: Appending PKG archive to EXE
12047 INFO: Fixing EXE headers
14497 INFO: Building EXE from EXE-00.toc completed successfully.
14497 INFO: checking COLLECT
14497 INFO: Building COLLECT because COLLECT-00.toc is non existent
14497 INFO: Building COLLECT COLLECT-00.toc
15561 INFO: Building COLLECT COLLECT-00.toc completed successfully.
15561 INFO: Build complete! The results are available in: C:\Users\deb\dist

C:\Users\deb>
```

Use file explorer to navigate to the location of the executable from the previous step

Left click on the file to select it

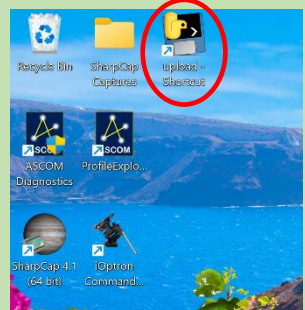
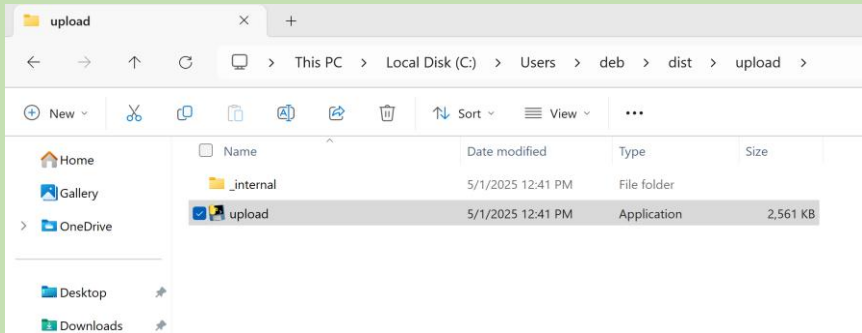
Right click to open Windows options

Goto "Send To" > Desktop to create shortcut

-You may have to select "More Options" to find "Send To" in menu

This will create a shortcut to the program on your Desktop

When opened, the upload executable should look like this.



```
upload - Shortcut
This program will monitor C:\DEB\Upload for new files and automatically upload them
In monitor mode

Cleaned up Upload folder to avoid unexpected behavior

|
```

Congratulations!! You have completed DEB upload configuration!

Appendix I – Update Existing Installation to 2026 Version

These instructions cover the steps to update an early installation of DEB to the newest version.

Purpose: Planetary System Stacker requires older Python and numpy that are not compatible with the new processing functions. Attempts at resolving these incompatibilities are proving unstable. To resolve this, DEB is moving to a 2 python installation.

Caveats: 1) These instructions use Python.org distributions, but users are welcome to use MiniConda3 (or any other preferred Python distribution). You are responsible for locating the proper MiniConda3 version and ensuring the proper package versions if you choose to use other than Python.org.

2) These instructions add the newest version of Python to the PATH variable. If you do not wish to use this method, or already have a configuration on your system you want to preserve, you are welcome to adjust the instructions as needed. It is assumed that, if this is a consideration for you, you know what you are doing well enough to handle the required changes to the instructions.

Download the newest files from: https://github.com/Chris-Mandrell/DEB_Scripts

Place new code in C:\DEB\Programs folder or wherever you have been running code from.

The following instructions are written to work from any command prompt. Section 1 of these instructions may be useful to you if you are unsure of how to open a command prompt.

1) (Not mandatory, but cleaner) Uninstall miniconda3 with one of the following commands:

```
winget uninstall miniconda3  
winget uninstall --id Anaconda.Miniconda3
```

2) Complete [Step 1a](#) and [Step 1b](#) in the instructions then return here to continue.

3) Install Planetary System Stacker using the following command (make sure you use the Python 3.8 installation!):

```
path\to\python38\python -m pip install planetary-system-stacker
```

4) Replace pss_console.py in planetary system stacker (PSS) with DEB version from scripts download -the location for your new PSS files will be:

```
path\to\python38\Lib\site-packages\planetary_system_stacker
```

9) Change the following 2 lines in deb.ini file in the [SITE] section (Note: use correct path... the ones below are likely options) (NOTE: deb.ini found in C:\Users\YourLoginName folder)

```
search_path = C:\Users\YourLoginName\AppData\Local\Programs\Python\Python38  
python_version = C:\Users\YourLoginName\AppData\Local\Programs\Python\Python38\python.exe
```

10) Create upload.py executable (first line below should be where you have your DEB programs)

```
cd C:\DEB\Programs  
python -m PyInstaller upload.py  
**if prompted to replace existing folder, enter y
```

- 11) Create shortcut on Desktop to upload.exe found in "C:\DEB\Programs\dist\upload" folder
 - left click on file to select
 - right click on file to open options
 - go to "Show more options"
 - go to "Send to" > "Desktop (create shortcut)"
- 12) Contact DEB coordinator for instructions on how to configure ssh upload